Minimally Sufficient Experimental Design Using Mathematical Modeling

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This was a two-person, two-semester research project at TCNJ, advised by Dr. Jana Gevertz, applying the minimally sufficient experimental design framework outlined in her paper [1]. Collecting experimental data can be expensive or invasive, so this framework identifies the least amount of data needed to still trust a model's predictions. We built a PK/PD model with 4 compartments and 7 parameters (drug distribution plus receptor binding) that describes how a drug circulates through the body and binds to receptors at its site of action, and used it to generate synthetic drug-concentration data. The goal of the project was to determine the minimum number of data collection days needed to confidently estimate the parameters in our model, and which days those should be. My partner and I developed the multistart optimization and profile likelihood code together.

Framework

Since the ODE model cannot be solved analytically, we solved it numerically using MATLAB's `ode45`. Rather than using real, noisy data, we extracted daily drug concentrations in the central compartment from the model solution to use as our simulated data, since this is the quantity that would actually be measured in an experiment. We then fit the model to the simulated data using MATLAB's `fmincon`, a gradient-based method, to find the parameter values that minimize the sum of squared errors (SSE), imposing bounds on each parameter based on biological considerations. Because the data was generated from the model itself, it is theoretically possible to achieve an SSE of 0, but in a 7-dimensional optimization problem the algorithm is likely to converge to a local minimum instead of the global one. To mitigate this, we developed a multistart algorithm that runs the optimization from 50 Sobol points. Finally, since very different sets of parameter values can fit the data almost equally well, we used the profile likelihood method to test which parameters could be reliably identified from the data. For each parameter, we fixed it at a grid of values and re-optimized the remaining free parameters; the best SSE at each value traces out the profile. A parameter is practically identifiable if its profile likelihood curve is a single valley that crosses the 90% confidence threshold (based on the chi-square distribution, measured relative to the best-fit SSE) on both sides of its minimum.

Results

Our best fit achieved an SSE of 7.6×10^{-10}, with every fitted parameter close to its true value. However, we also found a parameterization with an SSE of 1.98×10^{-7} whose values were far from both the best fit and the truth (for example, the volume ratio j was fit at 2.95 instead of 1), even though its curve was visually indistinguishable from the true one. The profile likelihood analysis confirmed the problem: even with daily measurements, none of the 7 parameters were practically identifiable, and the binding rates k_{on} and k_{off} had flat profiles, meaning they were not even structurally identifiable. Without identifiable parameters, we cannot trust the model's predictions, since parameter sets that describe the data equally well could make very different predictions.

Following the framework, we fixed the parameters that can be well-approximated from other experiments: the drug decay rate, the binding and dissociation rates, and the volume ratios. With these fixed and data collected daily, the two distribution rates, k_2 and k_3, became practically identifiable. We then asked how much of the daily data we actually need by testing protocols that sample 1, 3, 5, and 7 days out of a 20-day window. While k_2 became identifiable under many protocols with only a few measurements, k_3 was the bottleneck: the left side of its profile did not cross the confidence threshold until we reached 7 measurement days (Figure 1).

Seven days was the minimum, but the timing mattered just as much as the number. Out of 77,520 possible 7-day protocols, we checked every 10th protocol because it was computationally expensive, and only 52 (0.67%) resulted in both parameters being identifiable. These protocols followed a clear pattern: the first measurement must be taken on day 1 or 2, the second almost always on the following day, and then no further data is needed until day 9 or later (Figure 2). Early measurements capture the fast initial drop in concentration, while measurements from day 9 onward capture the slower phase driven by drug returning from the peripheral compartment; k_3 can only be determined when both phases are sampled. In short, our analysis gives experimentalists an actionable recommendation, 7 specific days instead of 20, while still being able to trust the model's parameterization and, therefore, its predictions.

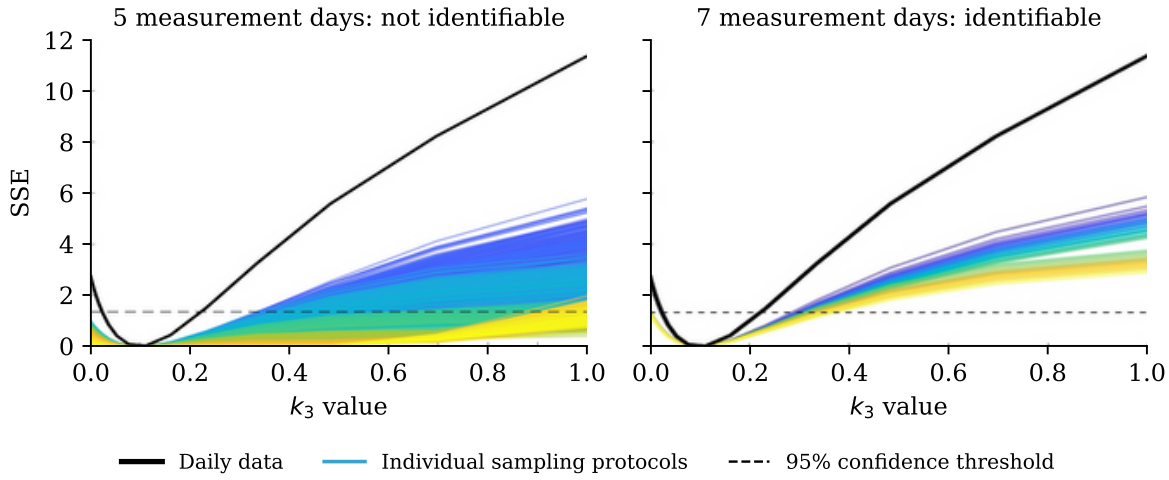


Figure 1: k_3 profile likelihood curves for 5-day (left) and 7-day (right) sampling protocols. Each colored curve is a different protocol, the black curve uses daily data, and the dashed line is the 90% confidence threshold. With 5 days, the left ends of the protocol curves stay below the threshold, so k_3 cannot be determined; with 7 days, they are (barely) above the threshold.

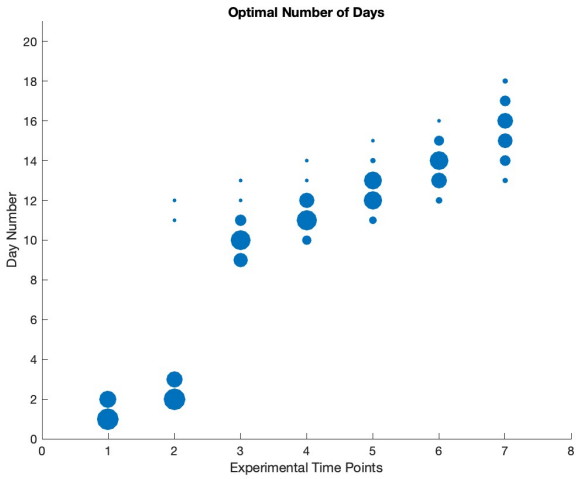


Figure 2: Collection days used by the 52 identifiable 7-day protocols, where bubble size shows the frequency of each day. Measure on day 1 or 2 and the following day, then wait until day 9 or later.

Limitations and Next Steps

Our simulated data was noise-free, so these results represent a best-case scenario. The original framework adds variability by sampling parameters from distributions, and found that noisier data reduces the number of protocols that preserve identifiability [1], so adding noise is the natural next step and would likely raise the number of required days. Our screen checked only the endpoints of each profile, and the fill 20-point profiles of the 52 passing protocols were checked visually, so next time we would automate that check and sample randomly instead of every 10th protocol. We also did not test whether the 52 identified protocols are robust to shifting a measurement by a single day, which matters if a lab misses a scheduled collection.

References

[1] Jana L. Gevertz and Irina Kareva. Minimally sufficient experimental design using identifiability analysis. *npj Systems Biology and Applications*, 10(1):2, 2024.